

SQL Insights Report

Querying the E-Commerce Orders Dataset

Data Analytics Assignment III

Overview

This report uses SQL to extract insights from the same 1,200-row e-commerce orders dataset used in the earlier EDA. The dataset was loaded into a relational table named orders (fields: OrderID, Date, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemsInCart, CouponCode, ReferralSource, TotalPrice). Ten queries were written to demonstrate core SQL skills such as SELECT, WHERE, ORDER BY, GROUP BY, HAVING, and aggregate functions (COUNT, SUM, AVG, MIN, MAX) with each query followed by its actual output and a short interpretation.

Queries and Results

Query 1: Top 5 delivered orders by value (SELECT, WHERE, ORDER BY)

SQL:

```
SELECT TOP 5
ORDERID,
PRODUCT,
QUANTITY,
UNITPRICE,
TOTALPRICE,
ORDERSTATUS
FROM DATA
WHERE ORDERSTATUS = 'DELIVERED'
ORDER BY TOTALPRICE DESC
```

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Result (top 5 shown):

ORDERID	PRODUCT	QTY	UNIT PRICE	TOTAL PRICE	ORDERSTATUS
ORD200789	Tablet	5	691.28	3456.40	Delivered
ORD200632	Laptop	5	678.16	3390.80	Delivered
ORD201065	Printer	5	666.80	3334.00	Delivered
ORD200361	Printer	5	659.85	3299.25	Delivered
ORD200837	Chair	5	655.55	3277.75	Delivered

Insight: the highest-value delivered orders are concentrated among Tablets, Laptops, Printers, and Chairs, all at or near maximum quantity (5) and maximum unit price consistent with the outliers identified in the earlier EDA.

Query 2: Revenue by product category (GROUP BY, SUM, AVG, COUNT)

SQL:

```
SELECT PRODUCT, COUNT(*) AS NUMORDERS, SUM(TOTALPRICE) AS
TOTALREVENUE,
AVG(TOTALPRICE) AS AVGORDERVALUE
FROM DATA
GROUP BY PRODUCT
ORDER BY TOTALREVENUE DESC;
```

Result:

PRODUCT	ORDERS	TOTAL REVENUE (\$)	AVG VALUE (\$)	ORDER
CHAIR	178	195,620.11	1,098.99	
PRINTER	181	195,612.61	1,080.73	
LAPTOP	173	192,126.56	1,110.56	
TABLET	179	186,568.95	1,042.28	
MONITOR	163	175,651.41	1,077.62	

DESK	170	167,459.93	985.06
PHONE	156	151,722.39	972.58

Insight: revenue is well balanced across categories. Chairs and Printers lead in total revenue despite Laptops having the highest average order value, meaning Laptop sales are fewer but individually larger.

Query 3: Average order value by payment method, excluding cancellations (WHERE + GROUP BY + AVG)

SQL:

```
SELECT PAYMENTMETHOD, COUNT(*) AS NUMORDERS,
       ROND(AVG(TOTALPRICE),2) AS AVGORDERVALUE
FROM DATA
WHERE ORDERSTATUS! = 'CANCELLED'
GROUP BY PAYMENTMETHOD
ORDER BY AVGORDERVALUE DESC;
```

Result:

PAYMENT METHOD	ORDERS	AVG ORDER VALUE (\$)
CREDIT CARD	180	1,109.79
CASH	197	1,034.95
DEBIT CARD	188	1,033.73
GIFT CARD	180	1,022.85
ONLINE	205	1,006.16

Insight: Among non-cancelled orders, Credit Card purchases have the highest average value (\$1,109.79), while Online payments are the most frequent but have the lowest average value.

Query 4: Order status distribution (GROUP BY, COUNT, computed percentage)

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SQL:

```
SELECT ORDERSTATUS, COUNT(*) AS NUMORDERS,  
ROUND (100.0 * COUNT(*) / (SELECT COUNT(*) FROM ORDERS), 2) AS  
PCTOFORDERS  
FROM DATA  
GROUP BY ORDERSTATUS  
ORDER BY NUMORDERS DESC;
```

Result:

ORDER STATUS	ORDERS	% OF ORDERS
CANCELLED	250	20.83%
RETURNED	247	20.58%
PENDING	237	19.75%
SHIPPED	235	19.58%
DELIVERED	231	19.25%

Insight: only 19.25% of orders are ultimately Delivered, while Cancelled and Returned together account for over 41% of all orders the most concerning pattern in the dataset.

Query 5: High-value orders by referral source (WHERE + GROUP BY + SUM)

SQL:

```
SELECT REFERRALSOURCE, COUNT(*) AS NUMORDERS, SUM(TOTALPRICE)  
AS TOTALREVENUE  
FROM DATA  
WHERE TOTALPRICE > 1000  
GROUP BY REFERRALSOURCE  
ORDER BY TOTALREVENUE DESC;
```

Result:

REFERRAL SOURCE	ORDERS >\$1000	TOTAL REVENUE (\$)
INSTAGRAM	113	207,008.41
FACEBOOK	107	195,622.89
EMAIL	103	194,628.46
GOOGLE	100	184,070.27
REFERRAL	89	162,230.04

Insight: Instagram drives the most high-value orders (>\$1,000) and the highest associated revenue, followed closely by Facebook and Email; direct Referral traffic contributes the least.

Query 6: Yearly order volume and revenue (GROUP BY on derived date part)

SQL:

```
SELECT
  YEAR([DATE]) AS ORDERYEAR,
  COUNT(*) AS NUMORDERS,
  SUM(TOTALPRICE) AS TOTALREVENUE
FROM DATA
GROUP BY YEAR([DATE])
ORDER BY YEAR([DATE])
```

Result:

ORDER YEAR	ORDERS	TOTAL REVENUE (\$)
2023	510	552,643.24
2024	459	480,235.87
2025	231	231,882.85

Insight: order volume and revenue declined from 2023 to 2024. 2025 figures cover only the first half of the year (through June), so they are not directly comparable to the full-year totals for 2023 and 2024.

Query 7: Price range by product (MIN, MAX, AVG)

```
SQL:
SELECT PRODUCT, MIN(UNITPRICE) AS MINPRICE, MAX(UNITPRICE) AS
MAXPRICE,
ROUND(AVG(UNITPRICE),2) AS AVGPRICE
FROM DATA
GROUP BY PRODUCT
ORDER BY AVGPRICE DESC;
```

Result:

PRODUCT	MIN PRICE (\$)	MAX PRICE (\$)	AVG PRICE (\$)
PHONE	11.39	695.10	375.22
TABLET	17.24	699.88	367.68
MONITOR	14.69	696.71	358.66
LAPTOP	18.20	699.93	357.71
CHAIR	14.93	695.29	355.66
PRINTER	12.10	697.19	351.71
DESK	26.95	697.93	329.61

Insight: every product spans a nearly identical price range (roughly \$11–\$700), so unit price is not a strong differentiator between categories Phones simply have the highest average price and Desks the lowest.

Query 8: Coupon usage and effectiveness (WHERE, GROUP BY, COUNT, AVG)

SQL:

```
SELECT      COUPONCODE,      COUNT(*)      AS      TIMESUSED,
ROUND(AVG(TOTALPRICE),2) AS AVGORDERVALUE
FROM DATA
WHERE COUPONCODE != 'UNKNOWN'
GROUP BY COUPONCODE
ORDER BY TIMESUSED DESC;
```

Result:

COUPON CODE	TIMES USED	AVG ORDER VALUE (\$)
FREESHIP	313	1,070.41
WINTER15	292	1,035.90
SAVE10	286	1,065.87

Insight: FREESHIP is the most frequently used coupon and is associated with the highest average order value, though the three coupons perform similarly overall none stands out as driving materially larger baskets.

Query 9: Order status breakdown for two products (WHERE with IN, GROUP BY multiple columns)

SQL:

```
SELECT PRODUCT, ORDERSTATUS, COUNT(*) AS NUMORDERS
FROM DATA
WHERE PRODUCT IN ('LAPTOP','PHONE')
GROUP BY PRODUCT, ORDERSTATUS
ORDER BY PRODUCT, NUMORDERS DESC;
```

Result:

PRODUCT	ORDER STATUS	ORDERS
LAPTOP	Delivered	40
LAPTOP	Returned	39
LAPTOP	Cancelled	35
LAPTOP	Pending	30
LAPTOP	Shipped	29
PHONE	Delivered	38
PHONE	Returned	31
PHONE	Cancelled	31
PHONE	Shipped	28
PHONE	Pending	28

Insight: both Laptops and Phones show 'Delivered' as their most common outcome individually, unlike the dataset overall where Cancelled edges out Delivered status distribution varies somewhat by product.

Query 10: Repeat customers (GROUP BY, HAVING, COUNT, SUM)

SQL:

```
SELECT TOP 5
CUSTOMERID,
COUNT(*) AS NUMORDERS,
SUM(TOTALPRICE) AS TOTALSPENT
FROM DATA
GROUP BY CUSTOMERID
HAVING COUNT(*) > 1
```

Result (top 5 shown):

CUSTOMER ID	ORDERS	TOTAL SPENT (\$)
C14847	2	1097.81

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C21191	2	647.77
C35852	2	1248.39
C38840	2	5723.23
C46651	2	1360.89

Insight: repeat customers (those with more than one order) are rare and each place at most 2 orders in this dataset, suggesting customer IDs are largely unique per transaction rather than reflecting a loyal repeat-buyer base.

Summary of Key Insights

- Revenue is evenly distributed across product categories (\$151.7K–\$195.6K), with no single product dominating sales.
- Only 19.25% of orders are Delivered; Cancelled (20.83%) and Returned (20.58%) are the two most common outcomes, together exceeding 41% of all orders.
- Credit Card orders carry the highest average value (\$1,109.79) among non-cancelled orders, while Online payments are most frequent but lowest in average value.
- Instagram is the top referral source for high-value orders (>\$1,000), ahead of Facebook and Email.
- Order volume and revenue trended downward from 2023 (\$552.6K) to 2024 (\$480.2K), with 2025 data covering only January–June.
- Coupon usage is broadly similar across FREESHIP, WINTER15, and SAVE10, with FREESHIP slightly ahead on both frequency and average order value.
- Repeat purchases are minimal nearly every customer in the dataset placed only a single order.

Conclusion

These SQL queries demonstrate core querying skills filtering with WHERE, sorting with ORDER BY, grouping with GROUP BY and HAVING, and aggregating with COUNT, SUM, AVG, MIN, and MAX while surfacing concrete business insights from the orders dataset. The most actionable finding is the low delivery-completion rate (19.25%) alongside high cancellation and return rates, which would be a natural focus for further investigation.

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